Comments on the code and some mathematical reasoning
Call Nr. 5 – Fixed version

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1 Introduction and Assumptions

$$\frac{\partial C}{\partial t} = -\frac{\partial F_x C}{\partial x} - \frac{\partial F_y C}{\partial y} + \frac{\partial^2 D_{xx} C}{\partial x^2} + \frac{\partial^2 D_{yy} C}{\partial y^2} + \frac{\partial^2 D_{xy} C}{\partial x \partial y}, \tag{1}$$

$$\mu = (m, q) \tag{2}$$
$$F_x(x,y) = \sum_{\mu \in \mathcal{M}} m \left(\beta_\mu(x,y) C_\mu - \alpha_\mu(x,y)\right) \tag{3}$$
$$F_y(x,y) = \sum_{\mu \in \mathcal{M}} q \left(\beta_\mu(x,y) C_\mu - \alpha_\mu(x,y)\right) \tag{4}$$
$$D_{xx}(x,y) = \frac{1}{2} \sum_{\mu \in \mathcal{M}} m^2 \left(\beta_\mu(x,y) C_\mu + \alpha_\mu(x,y)\right) \tag{5}$$
$$D_{yy}(x,y) = \frac{1}{2} \sum_{\mu \in \mathcal{M}} q^2 \left(\beta_\mu(x,y) C_\mu + \alpha_\mu(x,y)\right) \tag{6}$$
$$D_{xy}(x,y) = \sum_{\mu \in \mathcal{M}} mq \left(\beta_\mu(x,y) C_\mu + \alpha_\mu(x,y)\right). \tag{7}$$

$$\beta_{(1,0)}(x,y) = \beta_{(1,1)}(x,y) = x^{1/2} \tag{8}$$
$$\alpha_{(1,0)}(x,y) = \beta_{(1,0)} \exp\left(-E_{(1,0)}(x)\right) \tag{9}$$
$$\alpha_{(1,1)}(x,y) = \beta_{(1,1)} \exp\left(-E_{(1,1)}(x)\right), \tag{10}$$
$$\tag{11}$$

with

$$E_{(1,0)}(x) = 71.29 - 114.81(x^{2/3} - (x-1)^{2/3}) \tag{12}$$
$$E_{(1,1)}(x) = 77.56 - 114.81(x^{2/3} - (x-1)^{2/3}). \tag{13}$$

In our simplified model, we track the evolution of clusters in a 2D size space (x, y). The domain is restricted to $x \geq 0$ and $y \geq 0$. We make the fundamental assumption that the mobile clusters are restricted to the set $\mathcal{M} = \{(0,1),(1,1)\}$ not ($\mathcal{M} = \{(1,0),(1,1)\}$). We assume that the activation energies for these two mobile species are equal: $E_{(1,1)} = E_{(1,0)}$.

Our main focus is on the diffusion operator:

$$\text{Diffusion}(C) = \frac{\partial^2}{\partial x^2}(D_{xx} C) + \frac{\partial^2}{\partial y^2}(D_{yy} C) + \frac{\partial^2}{\partial x \partial y}(D_{xy} C) \tag{14}$$

Right now we focused on:

$$h = dx = dy.$$

$$u_{i,j}^n := C(x_i, y_j, t^n).$$

2 Coefficient Fields from Code

For $x \geq x_{\text{activation}}$:

$$\beta(x) = \sqrt{x}, \tag{15}$$

$$E(x) = 71.29 - 114.81 \left(x^{2/3} - (x-1)^{2/3}\right), \tag{16}$$

$$\alpha(x) = \beta(x) e^{-E(x)}, \tag{17}$$

$$\text{rate\_plus}(x) = \beta(x) c_m + \alpha(x), \tag{18}$$

$$d(x) = \max(\text{rate\_plus}(x), 0). \tag{19}$$

For $x < x_{\text{activation}}$, code sets $d(x) = 0$.
Inserted in the coefficient definitions:

$$D_{xx}(x,y) = \frac{1}{2} d(x), \tag{20}$$

$$D_{yy}(x,y) = d(x), \tag{21}$$

$$D_{xy}(x,y) = d(x). \tag{22}$$

Inserted back in the diffusion part of the PDE:

$$\frac{\partial^2}{\partial x^2}(D_{xx}C) + \frac{\partial^2}{\partial y^2}(D_{yy}C) + \frac{\partial^2}{\partial x \partial y}(D_{xy}C) = \frac{\partial^2}{\partial x^2}\left(\frac{d}{2}C\right) + \frac{\partial^2}{\partial y^2}(dC) + \frac{\partial^2}{\partial x \partial y}(dC). \tag{23}$$

All diffusion coefficients are therefore functions of the same scalar field $d(x)$.

3 Split Variables and Their Role

Solver split:

$$D_2 = \frac{1}{2} D_{xy}, \tag{24}$$

$$A_x = D_{xx} - D_2, \tag{25}$$

$$A_y = D_{yy} - D_2. \tag{26}$$

Interpretation:

$$A_x : \text{x-axis diffusion remainder}, \tag{27}$$

$$A_y : \text{y-axis diffusion remainder}, \tag{28}$$

$$D_2 : \text{mixed (diagonal) diffusion part}. \tag{29}$$

With $D_{xx} = d/2$, $D_{yy} = d$, $D_{xy} = d$:

$$D_2 = \frac{d}{2}, \tag{30}$$

$$A_x = \frac{d}{2} - \frac{d}{2} = 0, \tag{31}$$

$$A_y = d - \frac{d}{2} = \frac{d}{2}. \tag{32}$$

Diffusion decomposition:

$$\frac{\partial^2}{\partial x^2}(D_{xx}C) + \frac{\partial^2}{\partial y^2}(D_{yy}C) + \frac{\partial^2}{\partial x \partial y}(D_{xy}C) = \frac{\partial^2}{\partial y^2}(A_yC) + \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + 2\frac{\partial^2}{\partial x \partial y}\right)(D_2C). \tag{33}$$

Using

$$(\partial_{xx} + \partial_{yy} + 2\partial_{xy})\,\phi = (\partial_x + \partial_y)^2 \phi,$$

the last term is discretized by a diagonal second difference.

Why the discrete formula has the same three-point shape as axis terms:

$$s := \frac{x+y}{\sqrt{2}}, \tag{34}$$

$$\partial_x + \partial_y = \sqrt{2}\,\partial_s, \tag{35}$$

$$(\partial_x + \partial_y)^2 = 2\,\partial_{ss}. \tag{36}$$

Along the diagonal direction, one grid step changes (x, y) by (h, h), hence

$$\Delta s = \sqrt{2}\,h.$$

Therefore

$$2\,\partial_{ss}\phi \approx 2\,\frac{\phi_{i+1,j+1} - 2\phi_{i,j} + \phi_{i-1,j-1}}{(\Delta s)^2} = 2\,\frac{\phi_{i+1,j+1} - 2\phi_{i,j} + \phi_{i-1,j-1}}{2h^2} = \frac{\phi_{i+1,j+1} - 2\phi_{i,j} + \phi_{i-1,j-1}}{h^2}. \tag{37}$$

So the algebraic stencil form matches a centered second difference, but it is taken on diagonal neighbors and represents the mixed operator through $D_2 u$.

4 Axis and Diagonal Terms: Split Variable Formulas

For the diagonal solver, diffusion is assembled in conservative split flux form:

$$Q^x_{i+\frac{1}{2},j} = -A^x_{i+\frac{1}{2},j}\,\frac{u^n_{i+1,j} - u^n_{i,j}}{h} = 0 \text{ (In our simplified case where A}_x\text{=0)}, \tag{38}$$

$$Q^y_{i,j+\frac{1}{2}} = -A^y_{i,j+\frac{1}{2}}\,\frac{u^n_{i,j+1} - u^n_{i,j}}{h}, \tag{39}$$

$$Q^d_{i+\frac{1}{2},j+\frac{1}{2}} = -D^2_{i+\frac{1}{2},j+\frac{1}{2}}\,\frac{u^n_{i+1,j+1} - u^n_{i,j}}{h}, \tag{40}$$

with arithmetic averages for the interface coefficients. The split diffusion contribution is then

$$\text{diff}_{i,j} = -\frac{Q^x_{i+\frac{1}{2},j} - Q^x_{i-\frac{1}{2},j}}{h} - \frac{Q^y_{i,j+\frac{1}{2}} - Q^y_{i,j-\frac{1}{2}}}{h} - \frac{Q^d_{i+\frac{1}{2},j+\frac{1}{2}} - Q^d_{i-\frac{1}{2},j-\frac{1}{2}}}{h}. \tag{41}$$

$$\text{diff}_{i,j} = -\frac{Q^y_{i,j+\frac{1}{2}} - Q^y_{i,j-\frac{1}{2}}}{h} - \frac{Q^d_{i+\frac{1}{2},j+\frac{1}{2}} - Q^d_{i-\frac{1}{2},j-\frac{1}{2}}}{h}. \tag{42}$$

Inserting the equations for the Q in 42 gives again 33.

Where the diagonal stencil is re-found

In the final split formulation, the diagonal stencil is exactly the $\text{diff}_{\text{diag}}$ term above. It is the centered second difference taken on diagonal neighbors $(i \pm 1, j \pm 1)$.

Stencil kernel form:

$$\frac{1}{h^2}\begin{pmatrix} 0 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad \text{applied to } (D_2 u^n).$$

Substituting split variables:

$$D_2 = \frac{1}{2} D_{xy},$$

and for the current coefficient model $D_{xy} = d$,

$$D_2 = \frac{d}{2}, \qquad \text{diff}_{\text{diag}} = \frac{(du^n)_{i+1,j+1} - 2(du^n)_{i,j} + (du^n)_{i-1,j-1}}{2h^2}.$$

So the hinted diagonal stencil appears through the diagonal flux line $Q^d_{i+\frac{1}{2},j+\frac{1}{2}}$, i.e. through nearest diagonal pairs (i, j) and $(i+1, j+1)$.

5 Advection Discretization

For the standard and diagonal explicit baselines, advection is discretized in conservative upwind face-flux form:

$$v^x_{i+\frac{1}{2},j} = \frac{1}{2}\left(F_{x,i,j} + F_{x,i+1,j}\right), \tag{43}$$

$$v^y_{i,j+\frac{1}{2}} = \frac{1}{2}\left(F_{y,i,j} + F_{y,i,j+1}\right), \tag{44}$$

$$F^{\text{adv}}_{i+\frac{1}{2},j} = \begin{cases} v^x_{i+\frac{1}{2},j}\, u^n_{i,j}, & v^x_{i+\frac{1}{2},j} \geq 0, \\ v^x_{i+\frac{1}{2},j}\, u^n_{i+1,j}, & v^x_{i+\frac{1}{2},j} < 0, \end{cases} \tag{45}$$

$$F^{\text{adv}}_{i,j+\frac{1}{2}} = \begin{cases} v^y_{i,j+\frac{1}{2}}\, u^n_{i,j}, & v^y_{i,j+\frac{1}{2}} \geq 0, \\ v^y_{i,j+\frac{1}{2}}\, u^n_{i,j+1}, & v^y_{i,j+\frac{1}{2}} < 0, \end{cases} \tag{46}$$

and contributes through discrete divergence:

$$\text{adv}_{i,j} = -\frac{F^{\text{adv}}_{i+\frac{1}{2},j} - F^{\text{adv}}_{i-\frac{1}{2},j}}{h} - \frac{F^{\text{adv}}_{i,j+\frac{1}{2}} - F^{\text{adv}}_{i,j-\frac{1}{2}}}{h}. \tag{47}$$

This is basically a linear reduction of Godunov, ensuring that the same amount of mass enters and leaves grid-cells. Before implementing this, the regular upwinding introduced mass-drift.

6 Development Timeline Since the Last Meetings

The implementation path was iterative and is important for interpreting the final solver choices:

- First baseline: direct 9-point formulation (standard mixed-derivative stencil). It did not provide reliable behavior on the benchmark set.
- After the last meeting: we moved to the 5-point diagonal-split baseline. This worked much better, but the first attempts were still somewhat fragile in difficult cases.
- To improve advection handling, we implemented `strang_mp5`: Strang splitting with MP5 reconstruction and SSPRK3 for advection substeps.
- As the diagonal solver became robust, the Strang+MP5 path became less central for routine runs.
- We then implemented a Chang-Cooper variant but later removed it from active use because we did not find an explicit published method covering the diagonal/mixed term in the same formulation.

Please note, that we **do not** test this currently:

$$m_{(1,0)} = \iint C(x,y)x \,\mathrm{d}x\mathrm{d}y \tag{48}$$

$$m_{(0,1)} = \iint C(x,y)y \,\mathrm{d}x\mathrm{d}y. \tag{49}$$

But we test for general mass conservation right now:

$$m = \iint C(x,y) \,\mathrm{d}x\mathrm{d}y \tag{50}$$

7 Retained Method Set for Current Study

The active comparison now uses three solvers:

- standard_stencil: standard stencil baseline (explicit).
- diagonal_stencil: diagonal split stencil baseline (explicit).
- strang_mp5: Strang splitting with MP5 advection and diagonal diffusion.

8 Historical Note on Chang-Cooper

When researching solution approaches of Fokker-Planck, the Chang-Cooper method was frequently mentioned (example in [1], [2].) We implemented a Chang-Cooper variant during development as an additional conservative drift-diffusion method (followed [3] closely). However, after implementing it, we noticed that the paper never really discusses diagonal diffusion contributions. We also did not find any relevant source addressing this issue in a general manner online.

After getting the diagonal solver to work correctly, the necessity of another scheme was no longer present. Hence, Chang-Cooper was not continued.

9 Other promising ideas, directly tackling anisotropic

While trying to find Chang-Cooper anisotropic implementations we found these two papers. We don't know if they are applicable to our problem and if they would add value in the long-run. Can you please check them and tell us if we should invest time in them?

[4] is a very recent paper (2026). It offers a robust technique that extends the Chang-Cooper positivity guarantees to the problematic anisotropic cross-derivatives (Dxy). We are not sure if the case (Rosenbluth-Fokker-Planck) discussed in the paper includes our problem. L. Monasse: La référence [4] semble très intéressante, même si je pense que nous sommes ici dans un cas plus simple : on a bien une diffusion anisotrope, mais l'espace propre du null eigenspace est constant en fonction de l'espace. Cela simplifie sans doute l'utilisation d'un simple schéma IMEX au lieu de leur schéma EIN.

[5] talks about a local split step method. They claim strong performance gains. Interestingly, they explicitly talk about using Dxy, so this could maybe also be interesting. L. Monasse: La référence [5] est également intéressante, mais il faudrait comprendre plus en détails le splitting appliqué (je n'ai pas eu le temps de creuser).

References

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Appendix: Explicit Derivation of the Split Diffusion Operator

Assumptions

$$\mathcal{M} = \{(0,1),(1,1)\} \tag{51}$$

$$E_{(0,1)}(x) = E_{(1,1)}(x) \tag{52}$$

$$d(x,y) = \beta_{(0,1)}C_{(0,1)} + \alpha_{(0,1)} = \beta_{(1,1)}C_{(1,1)} + \alpha_{(1,1)} \tag{53}$$

Evaluating the Discrete Sums

$$D_{xx} = \frac{1}{2}\left[(0)^2 d(x,y) + (1)^2 d(x,y)\right] = \frac{1}{2}d(x,y) \tag{54}$$

$$D_{yy} = \frac{1}{2}\left[(1)^2 d(x,y) + (1)^2 d(x,y)\right] = d(x,y) \tag{55}$$

$$D_{xy} = [(0)(1)d(x,y) + (1)(1)d(x,y)] = d(x,y) \tag{56}$$

Operator Decomposition

To solve this numerically, we wish to extract the mixed derivative into a perfect square operator: $(\partial_x + \partial_y)^2 = \partial_{xx} + 2\partial_{xy} + \partial_{yy}$.

To match the coefficient of the ∂_{xy} cross-term, we must define a new split variable, D_2, such that:

$$2D_2 = D_{xy} \implies D_2 = \frac{1}{2}D_{xy} = \frac{1}{2}d(x,y) \tag{57}$$

$$A_x = D_{xx} - D_2 = \frac{1}{2}d(x,y) - \frac{1}{2}d(x,y) = 0 \tag{58}$$

$$A_y = D_{yy} - D_2 = d(x,y) - \frac{1}{2}d(x,y) = \frac{1}{2}d(x,y) \tag{59}$$

Final Formulation

$$\begin{aligned}
\mathrm{Diff}(C) &= \frac{\partial^2}{\partial x^2}(D_{xx}C) + \frac{\partial^2}{\partial y^2}(D_{yy}C) + \frac{\partial^2}{\partial x \partial y}(D_{xy}C) && (60)\\
&= \frac{\partial^2}{\partial x^2}\left((A_x + D_2)C\right) + \frac{\partial^2}{\partial y^2}\left((A_y + D_2)C\right) + \frac{\partial^2}{\partial x \partial y}(2D_2C) && (61)\\
&= \frac{\partial^2}{\partial x^2}(A_xC) + \frac{\partial^2}{\partial y^2}(A_yC) + \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + 2\frac{\partial^2}{\partial x \partial y}\right](D_2C) && (62)
\end{aligned}$$

Applying $A_x = 0$ and collapsing the perfect square:

$$\mathrm{Diff}(C) = \frac{\partial^2}{\partial y^2}\left(\frac{d(x,y)}{2}C\right) + (\partial_x + \partial_y)^2\left(\frac{d(x,y)}{2}C\right) \tag{63}$$